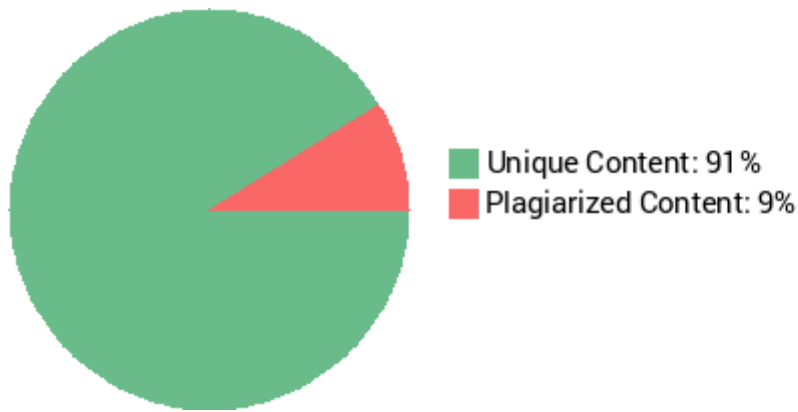




PLAGIARISM SCAN REPORT

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[Figure: System Architecture of Smart Grid Sentinel.]

The core of our predictive pipeline is the Informer model. It uses multi-head self-attention to map historical input sequences $X_{t-4:t}$ to future load shedding risk categories y_{t+1} . The model's hidden layers pass the attention representations through standard feed-forward networks, and a final Softmax classifier produces a probability distribution over the three target categories: Low, Medium, and High Risk.

A rule-based natural language layer interprets the output. Whenever the classifier predicts Medium or High Risk, the module evaluates the input variables against set thresholds (such as capacity overload boundaries or extreme THI values) to generate text explanations. This design provides immediate operational feedback without relying on complex deep-learning feature attribution maps.

Experimental Setup

We implemented all forecasting models in Python. Deep learning networks were trained in PyTorch and Tensor Flow/Keras on a local workstation utilizing an NVIDIA RTX graphics card. Each input sequence contains 5 historical steps (spanning 10 hours in total, with a 2-hour observation frequency). We trained the networks for 50 epochs using the Adam optimizer, setting the learning rate to 0.0005 and the batch size to 32. To evaluate performance, we measure classification Accuracy, Macro Precision, Macro Recall, Macro F1-score, and Weighted F1-score.

[Table: Hyperparameter Configurations]

<u>Parameter</u>	<u>Value</u>
---	---
Sequence Window Length	5
Training Epochs	50
Batch Size	32
Learning Rate	0.0005
Optimizer	Adam
Loss Function	Sparse Categorical Crossentropy

Results and Discussion

Quantitative Analysis

Table IV reports the quantitative evaluation metrics for all twelve models on the chronological test partition.

[Table: Performance Comparison of All Models]

Model	Acc	Macro P	Macro R	Macro F1	W. F1
---	---	---	---	---	---
Informer	87.41%	78.52%	80.26%	77.68%	88.44%
GRU	87.69%	79.00%	78.57%	77.53%	88.39%
Random Forest	87.38%	79.33%	79.97%	77.87%	88.34%
Times Net	87.04%	79.27%	80.65%	77.68%	88.19%
CNN-LSTM	87.16%	79.09%	79.47%	77.45%	88.13%
LSTM	86.85%	78.81%	79.10%	77.20%	87.82%
XGBoost	87.58%	77.78%	75.60%	76.23%	87.81%
Logistic Reg.	86.59%	78.30%	79.87%	77.05%	87.80%
Decision Tree	86.30%	78.66%	79.74%	76.88%	87.56%

PatchTST	85.69%	78.77%	80.49%	76.83%	87.22%
Autoformer	85.55%	78.00%	79.74%	76.36%	87.01%
TFT	84.38%	76.90%	78.77%	75.25%	86.06%

Informer Selection Justification

We selected the Informer model for production deployment over the GRU network, despite the latter's slightly higher raw accuracy (87.69% vs. 87.41%). The GRU model exhibited severe overfitting after 20 epochs, as evidenced by a diverging validation loss. In contrast, the Informer model converged stably across all 50 epochs (refer to Fig. 2). Furthermore, the Informer achieved a higher Weighted F1-score (88.44% compared to 88.39% for the GRU) and showed better predictive capability on the minority Medium-Risk class, reaching 79.00% recall compared to the GRU's 73.00%.

[Figure: Informer vs. GRU Training and Validation Loss Curves.]

Although the Random Forest model achieved a marginally higher Macro F1-score than the Informer (77.87% versus 77.68%, a difference of 0.19%), it remains unsuitable for actual grid deployment. Standard ensemble classifiers cannot capture sequential temporal dependencies or handle multi-step forecasting horizons without complex engineering. The Informer natively models chronological sequence dynamics through self-attention, making it much better suited for scale and multi-node grid expansions.

Training Time Complexity

Computational training efficiency is critical for localized, real-time grid deployment. Table V details the total training execution time for the key deep architectures over 50 epochs.

[Table: Training Time Comparison for Key Architectures]

The metrics indicate that the Informer trains highly efficiently. It executes approximately 3 times faster than PatchTST and Temporal Fusion Transformer (TFT) while maintaining strong predictive performance.

Error Analysis and Confusion Matrix

We analyze the specific classification patterns of the deployed Informer model using its

confusion matrix. Evaluated on the chronological test partition (10,027 samples), the confusion matrix is plotted in Fig. 3 and expressed mathematically below:

Equation: $C = \begin{bmatrix} 915 & 50 & 337 \\ 20 & 6970 & 620 \\ 75 & 160 & 880 \end{bmatrix}$

Here, rows and columns represent the True and Predicted risk categories respectively, ordered as [High, Low, Medium].

[Figure: Confusion Matrix for the PyTorch Informer Model.]

The Informer model correctly predicts 915 High-Risk, 6,970 Low-Risk, and 880 Medium-Risk test windows. Most classification errors occur between adjacent categories (such as classifying 337 High-Risk cases as Medium, or 620 Low-Risk cases as Medium). This is expected given that the underlying grid load and temperature transitions are continuous. More importantly, the model keeps critical false negatives very low, predicting High-Risk as Low-Risk in only 50 instances (3.84% of true High-Risk cases), which is essential for grid security and scheduling safety.

Intelligent Alert System & Rule-Based Explainability Layer

To facilitate smart grid operations, our architecture routes Informer classification outputs directly to a rule-based explainability layer. Instead of generating abstract deep learning attribution weights, this module checks input metrics against physical substation operating boundaries:

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